SPA617: RF Transmission, Reflections, and Standing Waves

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1 Aim of the Experiment

To understand signal propagation, reflection, and standing-wave formation in a coaxial transmission line at radio frequencies, and to experimentally evaluate fundamental cable parameters:

1. Determining the phase velocity (v_p) and velocity factor (VF) from standing wave voltage oscillations produced by open-circuit reflections.
2. Characterizing voltage reflection coefficients ($|\Gamma|$) and return loss (RL) under Open Circuit, Short Circuit, and Matched $50\,\Omega$ load terminations.
3. Estimating the line inductance (L') and capacitance (C') per unit length using measured phase velocity and line characteristic impedance $Z_0 \approx 50\,\Omega$.
4. Measuring insertion loss (IL) and attenuation per unit length (α) across a broad spectrum (10–1000 MHz) using a spectrum analyzer.

2 Theory and Principles

2.1 Transmission Line Waves and Round-Trip Phase

A coaxial cable guides Transverse Electromagnetic (TEM) waves. When an incident wave encounters a load impedance $Z_L \neq Z_0$, a fraction of the wave is reflected back toward the source. The complex voltage reflection coefficient at the load is:

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} \tag{1}$$

Special termination conditions include:

- **Open Circuit ($Z_L \to \infty$):** $\Gamma = +1$ (100% reflection, in-phase).
- **Short Circuit ($Z_L = 0$):** $\Gamma = -1$ (100% reflection, 180° out-of-phase).
- **Matched Load ($Z_L = Z_0$):** $\Gamma = 0$ (Zero reflection, maximum power transfer).

At a fixed observation point (the Tee junction), the oscilloscope registers the superposition of incident (V_{inc}) and reflected (V_{ref}) waves. As signal frequency f varies, the round-trip phase $\phi_{\text{RT}} = \frac{4\pi f l}{v_p} + \phi_\Gamma$ changes, producing periodic voltage amplitude oscillations between constructive maximum (V_{max}) and destructive minimum (V_{min}) node voltages:

$$V_{\text{max}} = |V_{\text{inc}}| + |V_{\text{ref}}|, \qquad V_{\text{min}} = |V_{\text{inc}}| - |V_{\text{ref}}| \tag{2}$$

2.2 Phase Velocity and Velocity Factor

Two consecutive voltage minima (or maxima) are separated by a round-trip phase change of 2π radians:

$$2\pi\Delta f\tau = 2\pi \implies \Delta f = \frac{1}{\tau} = \frac{v_p}{2l} \tag{3}$$

where l is the physical cable length and τ is the round-trip delay. Rearranging yields the phase velocity v_p and velocity factor VF:

$$v_p = 2l\Delta f \tag{4}$$

$$\mathrm{VF} = \frac{v_p}{c} \tag{5}$$

where $c = 3 \times 10^8\,\mathrm{m/s}$ is the speed of light in vacuum.

2.3 Reflection Coefficient and Return Loss

The magnitude of the reflection coefficient $|\Gamma|$ and Return Loss (RL) observed at the Tee junction are calculated as:

$$|\Gamma| = \frac{|V_{\mathrm{ref}}|}{|V_{\mathrm{inc}}|} = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}} \tag{6}$$

$$\mathrm{RL\ (dB)} = -20\log_{10}|\Gamma| \tag{7}$$

2.4 Distributed Parameters (L' and C')

For a low-loss TEM line with characteristic impedance $Z_0 \approx 50\,\Omega$ and phase velocity $v_p = 1/\sqrt{L'C'}$, the inductance per unit length (L') and capacitance per unit length (C') are:

$$L' = \frac{Z_0}{v_p}\ \ [\mathrm{H/m}], \qquad C' = \frac{1}{Z_0 v_p}\ \ [\mathrm{F/m}] \tag{8}$$

2.5 Cable Attenuation (α)

Cable insertion loss (IL) and attenuation per unit length (α) are given by:

$$\mathrm{IL\ (dB)} = P_{\mathrm{ref}}\ (\mathrm{dBm}) - P_{\mathrm{cable}}\ (\mathrm{dBm}) \tag{9}$$

$$\alpha = \frac{\mathrm{IL}}{l}\ \ [\mathrm{dB/m}] \tag{10}$$

3 Experimental Setup

Coaxial Transmission Line Measurement Setup:

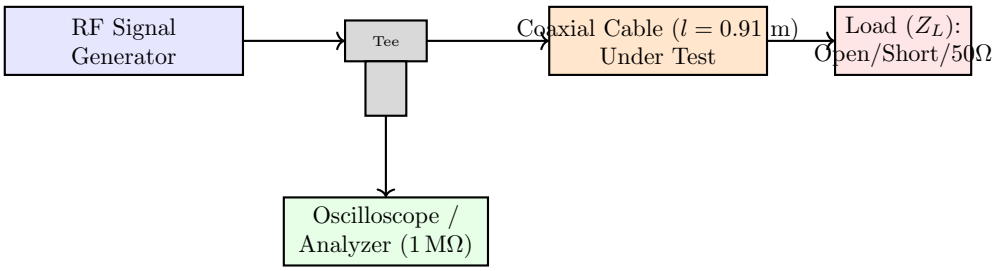


Figure 1: Experimental setup for standing-wave observation, reflection characterization, and cable attenuation.

4 Observation Tables and Quantitative Calculations

4.1 Part A: Phase Velocity and Velocity Factor Determination

Measured Cable Physical Length (l): 0.91 m

Table 1: Standing-wave voltage amplitude vs. frequency for open-ended coaxial cable

Frequency (MHz)	V_{rms}	**Frequency (MHz)**	V_{rms}	**Frequency (MHz)**	V_{rms}
3.0	1.98 V	80.0	1.48 V	160.0	578 mV
5.0	2.01 V	90.0	1.75 V	**170.0 (Min 2)**	**172 mV**
10.0	1.94 V	100.0	1.79 V	180.0	651 mV
20.0	1.60 V	110.0	1.72 V	190.0	1.30 V
30.0	1.19 V	120.0	1.54 V	200.0	1.57 V
40.0	744 mV	130.0	1.31 V	210.0	1.54 V
50.0	335 mV	140.0	1.08 V	220.0	1.35 V
60.0 (Min 1)	**174 mV**	150.0	840 mV	230.0	1.20 V
70.0	820 mV				

Quantitative Calculations for Part A: From the periodic standing-wave voltage curve, two consecutive resonance minima are observed at:

$$f_1 = 60.0 \text{ MHz} \quad (V_{\text{min1}} = 174 \text{ mV}), \qquad f_2 = 170.0 \text{ MHz} \quad (V_{\text{min2}} = 172 \text{ mV})$$

$$\Delta f = |f_2 - f_1| = 170.0 \text{ MHz} - 60.0 \text{ MHz} = \mathbf{110.0} \text{ MHz} = 1.10 \times 10^8 \text{ Hz} \tag{11}$$

$$v_p = 2l\Delta f = 2 \times 0.91 \text{ m} \times 1.10 \times 10^8 \text{ Hz} = \mathbf{2.002 \times 10^8} \text{ m/s} \tag{12}$$

$$\text{VF} = \frac{v_p}{c} = \frac{2.002 \times 10^8}{3.000 \times 10^8} = \mathbf{0.6673} \approx 0.67 \tag{13}$$

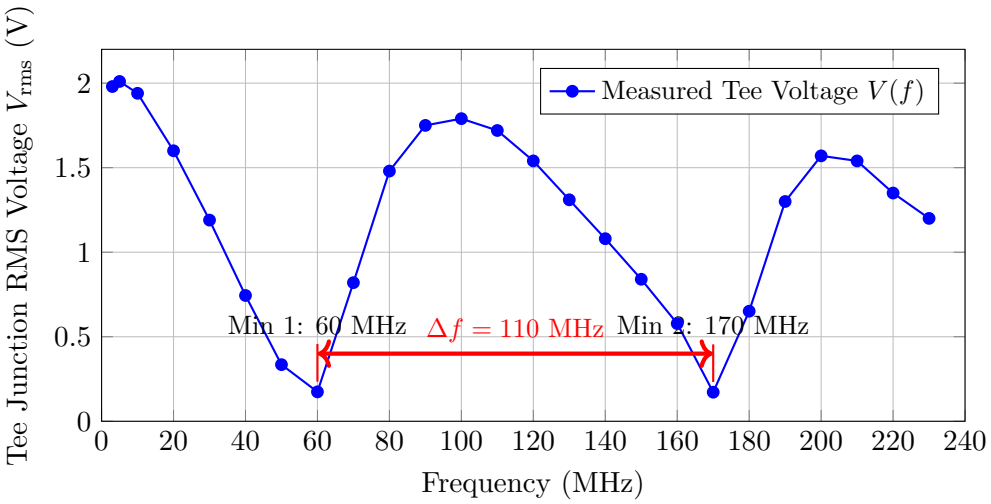


Figure 2: Standing-wave voltage amplitude oscillations at the Tee junction due to open-circuit round-trip phase interference.

4.2 Part B: Reflection Coefficients and Impedance Matching

Table 2: Measured standing wave envelope voltage limits, reflection coefficients, and return loss

Termination (Z_L)	$V_{\max}$ (mV)	$V_{\min}$ (mV)	V_{inc} (mV)	V_{ref} (mV)	$\lvert\Gamma\rvert$	**Return Loss RL (d**
Open Circuit ($Z_L \to \infty$)	920.0	34.0	477.0	443.0	**0.9287**	**0.64**
Short Circuit ($Z_L = 0$)	988.0	76.0	532.0	456.0	**0.8571**	**1.34**
Matched Load ($Z_L = 50\,\Omega$)	690.0	676.0	683.0	7.0	**0.0102**	**39.79**

Quantitative Calculations for Part B:

1. **Open Circuit Termination:**

$$V_{\text{inc}} = \frac{920 + 34}{2} = 477.0 \text{ mV}, \qquad V_{\text{ref}} = \frac{920 - 34}{2} = 443.0 \text{ mV} \tag{14}$$

$$|\Gamma_{\text{open}}| = \frac{920 - 34}{920 + 34} = \frac{886}{954} = \mathbf{0.9287} \approx 0.93 \tag{15}$$

$$\text{RL}_{\text{open}} = -20 \log_{10}(0.9287) = \mathbf{0.64} \text{ dB} \tag{16}$$

2. **Short Circuit Termination:**

$$V_{\text{inc}} = \frac{988 + 76}{2} = 532.0 \text{ mV}, \qquad V_{\text{ref}} = \frac{988 - 76}{2} = 456.0 \text{ mV} \tag{17}$$

$$|\Gamma_{\text{short}}| = \frac{988 - 76}{988 + 76} = \frac{912}{1064} = \mathbf{0.8571} \approx 0.86 \tag{18}$$

$$\text{RL}_{\text{short}} = -20 \log_{10}(0.8571) = \mathbf{1.34} \text{ dB} \tag{19}$$

3. **Matched $50\,\Omega$ Termination:**

$$V_{\text{inc}} = \frac{690 + 676}{2} = 683.0 \text{ mV}, \qquad V_{\text{ref}} = \frac{690 - 676}{2} = 7.0 \text{ mV} \tag{20}$$

$$|\Gamma_{50}| = \frac{690 - 676}{690 + 676} = \frac{14}{1366} = \mathbf{0.0102} \tag{21}$$

$$\text{RL}_{50} = -20 \log_{10}(0.0102) = \mathbf{39.79} \text{ dB} \tag{22}$$

Physical Match Inference: Connecting the $50\,\Omega$ load reduces the reflection coefficient from $|\Gamma_{\text{open}}| = 0.9287$ down to $|\Gamma_{50}| = 0.0102$ (a 39.79 dB return loss). This confirms that the transmission line is closely matched to $50\,\Omega$, proving its characteristic impedance $Z_0 \approx 50\,\Omega$.

4.3 Part C: Line Inductance and Capacitance Per Unit Length

Using the measured phase velocity $v_p = 2.002 \times 10^8$ m/s and characteristic impedance $Z_0 = 50\,\Omega$:

$$L' = \frac{Z_0}{v_p} = \frac{50}{2.002 \times 10^8} = 2.4975 \times 10^{-7} \text{ H/m} = \mathbf{249.75} \text{ nH/m} \tag{23}$$

$$C' = \frac{1}{Z_0 v_p} = \frac{1}{50 \times 2.002 \times 10^8} = 9.990 \times 10^{-11} \text{ F/m} = \mathbf{99.90} \text{ pF/m} \tag{24}$$

4.4 Part D: Cable Insertion Loss and Attenuation

Reference Input Power ($P_{\mathbf{ref}}$): 0.00 dBm (constant) **Cable Length (l):** 0.91 m

Table 3: Frequency-dependent power transmission, insertion loss, and attenuation per unit length

Frequency (MHz)	P_{ref} (dBm)	P_{cable} (dBm)	**Insertion Loss IL (dB)**	**Attenuation α (dB/m)**
10.0	0.0	-1.77	1.77	**1.945**
20.0	0.0	-1.88	1.88	**2.066**
30.0	0.0	-1.97	1.97	**2.165**
40.0	0.0	-2.06	2.06	**2.264**
50.0	0.0	-2.07	2.07	**2.275**
60.0	0.0	-2.02	2.02	**2.220**
70.0	0.0	-1.94	1.94	**2.132**
80.0	0.0	-1.88	1.88	**2.066**
90.0	0.0	-1.90	1.90	**2.088**
100.0	0.0	-2.04	2.04	**2.242**
150.0	0.0	-2.11	2.11	**2.319**
200.0	0.0	-2.53	2.53	**2.780**
250.0	0.0	-2.01	2.01	**2.209**
300.0	0.0	-2.39	2.39	**2.626**
350.0	0.0	-2.61	2.61	**2.868**
400.0	0.0	-2.47	2.47	**2.714**
450.0	0.0	-2.79	2.79	**3.066**
500.0	0.0	-2.76	2.76	**3.033**
700.0	0.0	-2.76	2.76	**3.033**
1000.0	0.0	-3.40	3.40	**3.736**

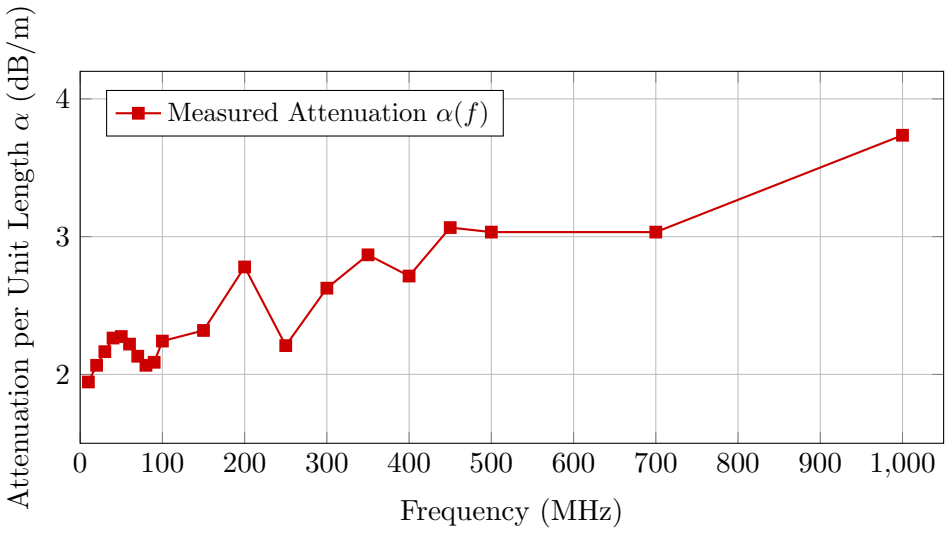


Figure 3: Cable attenuation per unit length (α) vs. frequency exhibiting dielectric and skin-effect losses.

5 Summary of Experimental Results

Table 4: Principal Measured and Derived Transmission Line Parameters

Parameter / Quantity	**Symbol**	**Measured / Calculated Value**
Cable Physical Length	l	0.91 m
Resonant Frequency Spacing	Δf	110.0 MHz
Signal Phase Velocity	v_p	2.002×10^8 m/s
Cable Velocity Factor	$\text{VF} = v_p/c$	$0.6673 \approx 0.67$
Reflection Coeff. (Open Circuit)	$\lvert\Gamma_{\text{open}}\rvert$	0.9287
Return Loss (Open Circuit)	RL_{open}	0.64 dB
Reflection Coeff. (Short Circuit)	$\lvert\Gamma_{\text{short}}\rvert$	0.8571
Return Loss (Short Circuit)	RL_{short}	1.34 dB
Reflection Coeff. (50 Ω Load)	$\lvert\Gamma_{50}\rvert$	0.0102
Return Loss (50 Ω Load)	RL_{50}	39.79 dB
Characteristic Impedance	Z_0	$\approx 50\,\Omega$ (Verified)
Inductance per Unit Length	L'	249.75 nH/m
Capacitance per Unit Length	C'	99.90 pF/m
Attenuation at 100 MHz	α_{100}	2.242 dB/m
Attenuation at 1000 MHz	α_{1000}	3.736 dB/m

6 Conclusions and Discussion

In this experiment, transmission line propagation and reflection phenomena were evaluated:

1. **Standing-Wave Oscillations & Velocity Factor:** Frequency sweeping with an open-ended cable produced voltage oscillations with a resonance spacing $\Delta f = 110.0$ MHz. This yielded a phase velocity $v_p = 2.002 \times 10^8$ m/s and Velocity Factor VF = 0.6673. This closely matches standard polyethylene-dielectric RG-58 coaxial cable specs (VF $\approx$ 0.66).

2. **Impedance Matching & Reflections:** Open and short circuit terminations exhibited near-total reflections ($|\Gamma_{\text{open}}| = 0.9287$, $|\Gamma_{\text{short}}| = 0.8571$). Connecting a 50 Ω load suppressed reflections down to $|\Gamma_{50}| = 0.0102$ (RL = 39.79 dB), confirming that $Z_0 \approx 50\,\Omega$.

3. **Line Parameters (L', C'):** Distributed inductance $L' = 249.75$ nH/m and capacitance $C' = 99.90$ pF/m were derived.

4. **Attenuation Profile:** Cable attenuation increased with frequency from 1.945 dB/m at 10 MHz to 3.736 dB/m at 1000 MHz due to conductor skin effect ($\propto \sqrt{f}$) and dielectric dissipation loss ($\propto f$).